CS1231S Discrete Structures

WEEK 10 RECITATION
LECTURE 11. COUNTING AND PROBABILITY II

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r-Combinations

Definition: *r*-combination

Let n and r be non-negative integers with $r \leq n$.

An ***r*-combination** of a set of n elements is a subset of r of the n elements.

$\binom{n}{r}$, read "n choose r", denotes the number of subsets of size r (r-combinations) that can be chosen from a set of n elements.

Other symbols used are $C(n, r)$, ${}_nC_r$, $C_{n,r}$, or ${}^nC_r$.

r-Combinations

Example 1: Let S = {Ann, Bob, Cyd, Dan}. Each committee consisting of three of the four people in S is a 3-combination of S.

a. List all such 3-combinations of S.

The 3-combinations are:

{Bob, Cyd, Dan}, {Ann, Cyd, Dan},
{Ann, Bob, Dan}, {Ann, Bob, Cyd}.

b. What is $\binom{4}{3}$? **4**

r-Combinations

Example 2: Ordered and unordered selections

Ordered selection	**Unordered selection**
Also called *r*-permutation	Also called *r*-combination

Example: $S = \{ 1, 2, 3 \}$

2-permutations of S	2-combinations of S
(1, 2) (2, 1)	{1, 2}
(1, 3) (3, 1)	{1, 3}
(2, 3) (3, 2)	{2, 3}

r-Combinations

Example 3: Relation between Permutations and Combinations

Write all 2-permutations of the set {0, 1, 2, 3}. Find an equation relating the number of 2-permutations, $P(4, 2)$, and the number of 2-combinations, $\binom{4}{2}$, and solve this equation for $\binom{4}{2}$.

(0, 1), (1, 0),
(0, 2), (2, 0),
(0, 3), (3, 0),
(1, 2), (2, 1),
(1, 3). (3, 1).
(2, 3), (3, 2)

According to Theorem 9.2.3,

$P(4, 2) = 4!/(4-2)! = 4!/2! = 12$

The construction of a 2-permutation of {0, 1, 2, 3} can be thought of comprising two steps:

Step 1: Choose a subset of 2 elements from {0, 1, 2, 3}.
Step 2: Choose an ordering for the 2-element subset.

Theorem 9.2.3:

$$P(n, r) = \frac{n!}{(n-r)!}$$

r-Combinations

Example 3: Relation between Permutations and Combinations

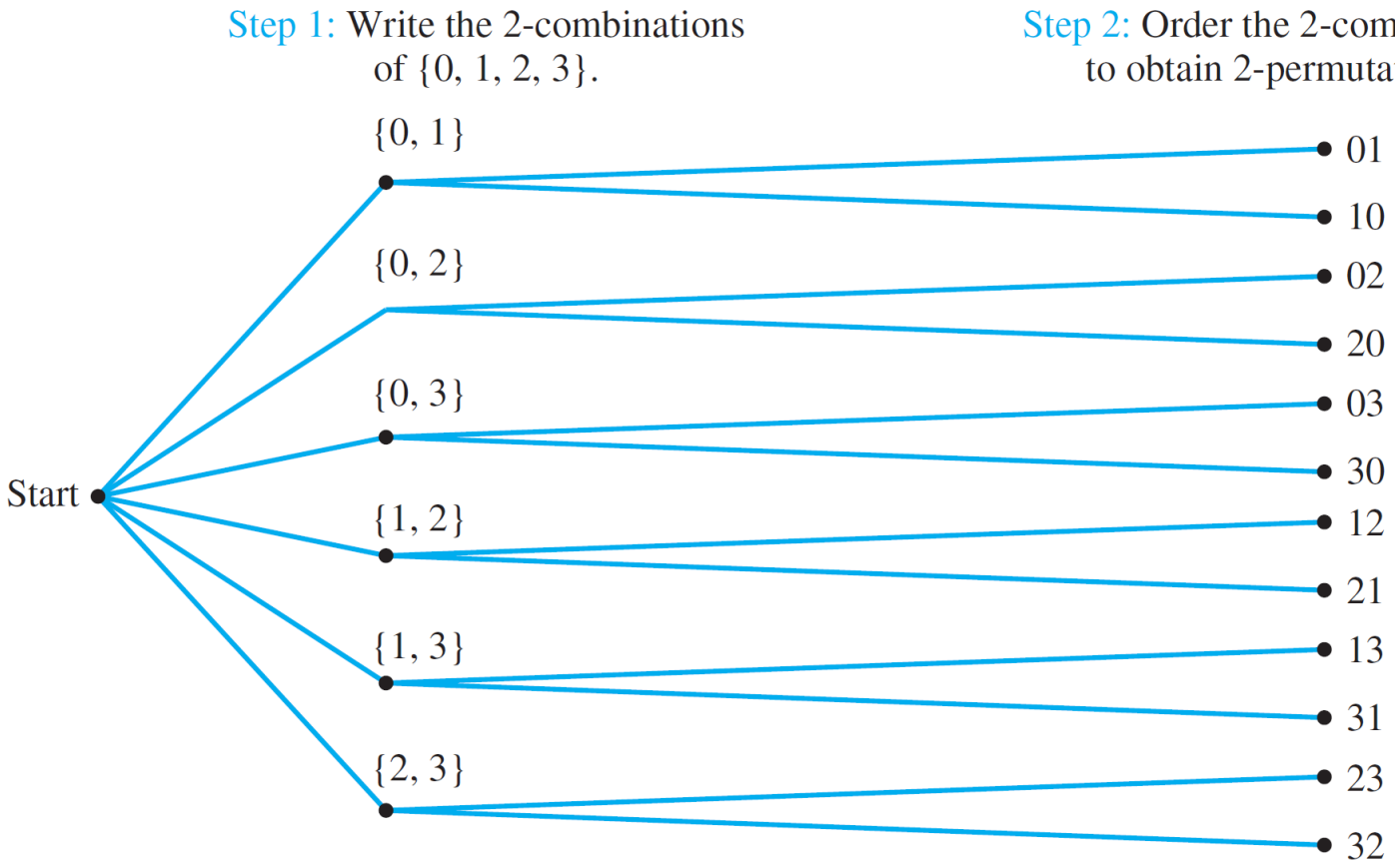


Figure 9.5.1 Relationship between Permutations and Combinations

The number of ways to perform step 1 is $\binom{4}{2}$.

The number of ways to perform step 2 is 2!

$$P(4, 2) = \binom{4}{2} \cdot 2!$$

$$\binom{4}{2} = P(4, 2) / 2!$$
$$= 12 / 2 = 6$$

r-Combinations

Theorem 9.5.1 Formula for $\binom{n}{r}$

The number of subsets of size *r* (or *r*-combinations) that can be chosen from a set of *n* elements, $\binom{n}{r}$, is given by the formula

$$\binom{n}{r} = \frac{P(n,r)}{r!}$$

or, equivalently,

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

where *n* and *r* are non-negative integers with $r \leq n$.

Recall Theorem 9.2.3:

$$P(n,r) = \frac{n!}{(n-r)!}$$

r-Combinations

Example 4: Suppose the group of 12 consists of 5 men and 7 women.

a. How many 5-person teams can be chosen that consist of 3 men and 2 women?

$$\binom{5}{3}\times\binom{7}{2}=\frac{5!}{3!\,2!}\times\frac{7!}{2!\,5!}=\mathbf{210}$$

b. How many 5-person teams contain at least one man?

Let A be the set of all 5-person teams, and B_k be the set of 5-person teams with k men, $0 \le k \le 5$.

$$|A|=\binom{12}{5}=792\,;\,|B_0|=\binom{7}{5}=21$$

$$792-21=\mathbf{771}$$

c. How many 5-person teams contain at most one man?

$$|B_1|=\binom{5}{1}\times\binom{7}{4}=5\times35=175$$

$$|B_0|+|B_1|=21+175=\mathbf{196}$$

Use difference rule or sum rule as appropriate.

Set with indistinguishable elements

Example 5: Order the letters in the word: **M I S S I S S I P P I**

How many distinguishable orderings are there?

Four-step process:

Step 1: Choose a subset of 4 positions for the S's. $\binom{11}{4}$

Step 2: Choose a subset of 4 positions for the I's. $\binom{7}{4}$

Step 3: Choose a subset of 2 positions for the P's. $\binom{3}{2}$

Step 4: Choose a subset of 1 position for the M. $\binom{1}{1}$

$$\binom{11}{4} \times \binom{7}{4} \times \binom{3}{2} \times \binom{1}{1} = \mathbf{34650}$$

Set with indistinguishable elements

Theorem 9.5.2 Permutations with Sets of Indistinguishable Objects

Suppose a collection consists of n objects of which

n_1 are of type 1 and are indistinguishable from each other
n_2 are of type 2 and are indistinguishable from each other
:
n_k are of type k and are indistinguishable from each other

and suppose that $n_1 + n_2 + \ldots + n_k = n$. Then the number of distinguishable permutations of the n objects is

$$\binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3}\cdots\binom{n-n_1-n_2-\cdots-n_{k-1}}{n_k} = \frac{n!}{n_1!n_2!n_3!\cdots n_k!}$$

Set with indistinguishable elements

Theorem 9.5.1:

$$\binom{n}{r} = \frac{n!}{r!\,(n-r)!}$$

Theorem 9.5.2 Permutations with Sets of Indistinguishable Objects

$$\binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3}\cdots\binom{n-n_1-n_2-\cdots-n_{k-1}}{n_k} = \frac{n!}{n_1!n_2!n_3!\cdots n_k!}$$

$$\binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3}\cdots\binom{n-n_1-n_2-\cdots-n_{k-1}}{n_k}$$

$$= \frac{n!}{n_1!\,(n-n_1)!} \times \frac{(n-n_1)!}{n_2!\,(n-n_1-n_2)!} \times \frac{(n-n_1-n_2)!}{n_3!\,(n-n_1-n_2-n_3)!} \times \cdots \times \frac{(n-n_1-n_2-\cdots n_{k-1})!}{n_k!\,(n-n_1-n_2-\cdots n_{k-1}-n_k)!}$$

$$= \frac{n!}{n_1!\,n_2!\,n_3!\cdots n_k!\,(n-n_1-n_2-\cdots n_{k-1}-n_k)!} = \frac{n!}{n_1!\,n_2!\,n_3!\cdots n_k!\,0!} = \frac{n!}{n_1!\,n_2!\,n_3!\cdots n_k!}$$

Exercises



Exercise 1: A set of 5 distinct integers is selected from $S = \{1,2,3,\cdots 12\}$.

(a) How many selections contain exactly 2 even integers?

$$\binom{6}{2} \times \binom{6}{3} = 15 \times 20 = \mathbf{300}$$

(b) How many selections contain at least one multiple of 3?

There are 4 multiples of 3 and 8 non-multiples of 3.

$$\binom{12}{5} - \binom{8}{5} = 792 - 56 = \mathbf{736}$$

Exercises



Exercise 2: A bookshelf contains 6 mathematics book, 5 computer science books, and 4 physics books. All books are distinct. How many ways are there to select 7 books such that at least 2 books from each subject are selected?

Identify the possible distributions:

2 books from each subject, the 7th book can be any subject.

3 cases: (1) 3 M, 2 CS, 2 P; (2) 2 M, 3 CS, 2 P; (3) 2 M, 2 CS, 3 M.

$$\binom{6}{3}\binom{5}{2}\binom{4}{2} = 20\times10\times6 = 1200$$

$$\binom{6}{2}\binom{5}{3}\binom{4}{2} = 15\times10\times6 = 900$$

$$\binom{6}{2}\binom{5}{2}\binom{4}{3} = 15\times10\times4 = 600$$

$$1200 + 900 + 600 = \mathbf{2700}$$

Exercises



Exercise 3: A box contains 6 red balls, 5 blue balls and 4 green balls. Balls of the same colour are identical. How many distinct arrangements are possible if exactly 3 red balls are selected and at least one ball of each of the other two colours is selected to get 7 balls arranged?

Similar to Exercise 2, but we can try another approach.

First, choose the 3 positions of the red balls: $\binom{7}{3}$

Remaining 4 positions: 2^4

But we must exclude BBBB and GGGG.

$$\binom{7}{3}(2^4 - 2) = 35 \times 14 = \mathbf{490}$$

Exercises



Exercise 4: How many distinct arrangements of the letters of BANANA are possible if the first and last letters must both be A?

After fixing the first and last positions, the four remaining letters are A,B,N,N.

$n = 4, n_1 = 1, n_2 = 1, n_3 = 2.$

Apply Theorem 9.5.2:

$$\frac{4!}{1!\,1!\,2!} = \frac{24}{2} = \mathbf{12}$$

Exercises



Exercise 5: How many distinct arrangements of the letters of MISSISSIPPI are possible if all four S's must appear consecutively?

Treat the 4 S's as a single block SSSS: (SSSS), M,I,I,I,I,P,P

$n = 8, n_1 = 1, n_2 = 1, n_3 = 4, n_4 = 2.$

Apply Theorem 9.5.2:

$$\frac{8!}{1!\,1!\,4!\,2!} = \frac{40320}{24\times 2} = \mathbf{840}$$

r-Combination with repetition allowed

At a hawker centre there are 6 stalls selling local delights. If you are buying lunch for 20 guests, how many different selections can you make?

Chicken rice



Nasi lemak



Mee rebus





Ayam penyet



Laksa



Bak chor mee

r-Combination with repetition allowed

How many ways are there to choose *r* elements without regard to order from a set of *n* elements if *repetition is allowed*?

Definition: Multiset

An ***r*-combination with repetition allowed**, or **multiset of size *r***, chosen from a set *X* of *n* elements is an unordered selection of elements taken from *X* with repetition allowed.

If $X = \{x_1, x_2, \cdots, x_n\}$, we write an *r*-combination with repetition allowed as $\left[x_{i_1}, x_{i_2}, \cdots, x_{i_r}\right]$ where each x_{i_j} is in *X* and some of the x_{i_j} may equal each other.

r-Combination with repetition allowed

Example 6: Write a complete list to find the multisets of size 3 that can be selected from {1, 2, 3, 4}.

All combinations with 1, 1:	[1, 1, 1]; [1, 1, 2]; [1, 1, 3]; [1, 1, 4]
All additional combinations with 1, 2:	[1, 2, 2]; [1, 2, 3]; [1, 2, 4]
All additional combinations with 1, 3:	[1, 3, 3]; [1, 3, 4]
All additional combinations with 1, 4:	[1, 4, 4]
All additional combinations with 2, 2:	[2, 2, 2]; [2, 2, 3]; [2, 2, 4]
All additional combinations with 2, 3:	[2, 3, 3]; [2, 3, 4]
All additional combinations with 2, 4:	[2, 4, 4]
All additional combinations with 3, 3:	[3, 3, 3]; [3, 3, 4]
All additional combinations with 3, 4:	[3, 4, 4]
All additional combinations with 4, 4:	[4, 4, 4]

20 3-combinations with repetition allowed.

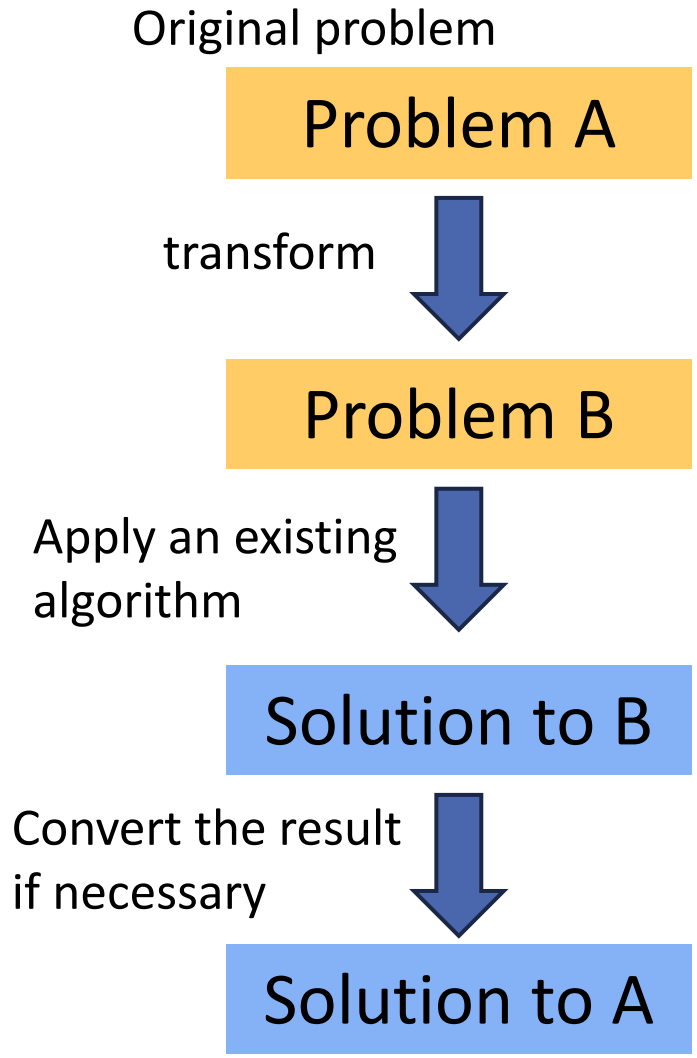
Problem Reduction

Problem reduction is a fundamental problem-solving technique in computer science in which we transform one problem into another problem whose solution we already know how to obtain.

> Instead of solving a new problem directly, transform it into a problem we already know how to solve, then use that solution to solve the original problem.

Problem reduction is important in algorithm design, computational complexity, computability theory, and artificial intelligence.

Problem Reduction



We say that **Problem A is reducible to Problem B**, written informally as $A \leq B$.

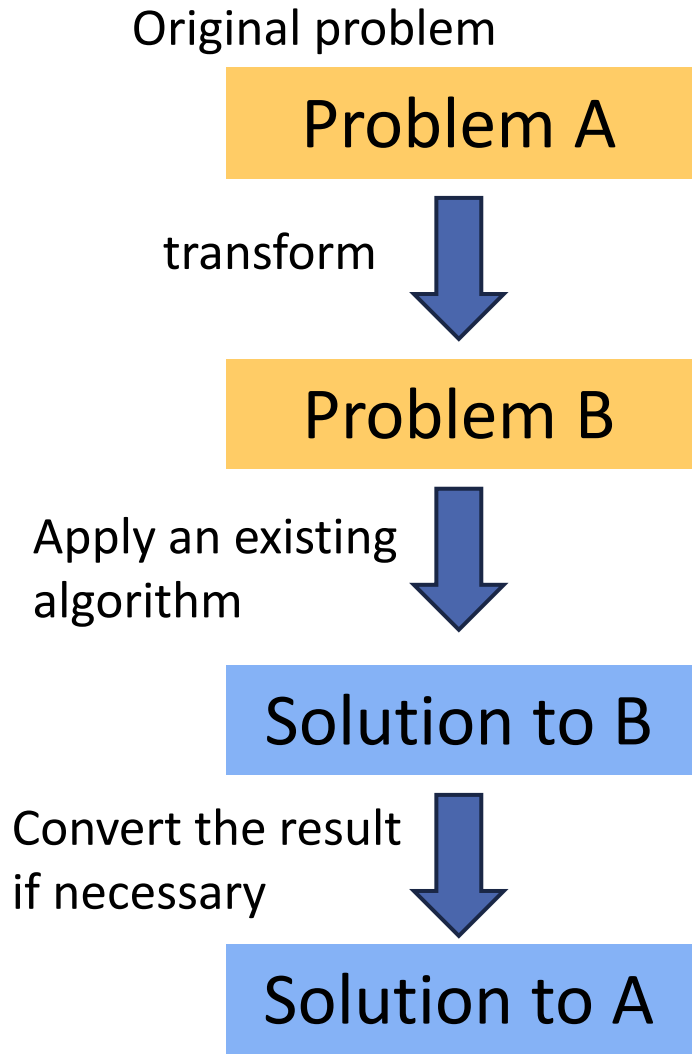
Eg: Finding the maximum by using a sort algorithm.

- Problem A: Find the maximum element.
- Problem B: Sort the array into ascending order.
- We can solve A by reducing it to B:
  - Sort the array into ascending order.
  - Return the last element of the sorted array.

We have solved the maximum-finding problem using a sorting algorithm. However, this reduction is not necessarily efficient.

- Finding the max directly: O(n)
- Using comparison sort: O(n log n)

Problem Reduction



Note: Problem reduction does not necessarily produce the most efficient algorithm. It allows us to reuse an existing solution.

Two purposes of reduction:

Purpose	Direction of reasoning
Algorithm design	We know how to solve B, so reduce A to B to obtain an algorithm for A.
Complexity theory	We know A is hard, so reduce A to B to establish that B is at least as hard as A.

Problem Reduction

A difficult problem may become easier when we recognise it as a familiar problem in disguise.

Just for Laugh #1



The Computer Science Professor and the Pantry Fire.

A professor is asked to solve two problems.

Problem 1: There is a fire in the pantry. A pail filled with water is standing nearby. What should you do?

Professor: “Simple! Pick up the pail and pour the water onto the fire.”

Problem 2: There is a fire in the pantry. This time, the pail is empty, but a tap is nearby. What should you do?

Professor: “Even simpler! Fill the pail with water. Then we have reduced the problem to Problem 1, which we already know how to solve!”

The professor is then given a third problem:

Problem 3: There is no fire in the pantry, and the pail is already full of water. What should you do?

Professor: “Easy! Start a fire in the pantry. Then we have reduced the problem to Problem 1!”

Just for Laugh #2



Professor: “How would you solve a problem that is too difficult for you?”

Student: “Easy! I reduce it to a problem I already know how to solve.”

Professor: “Excellent! Can you give me an example?”

Student: “Sure. I reduce my difficult assignment to a simpler problem: finding a classmate who has already solved it!”

Professor: “That’s not problem reduction. That’s academic misconduct!”

Student: “But Professor, isn’t the whole idea of problem reduction to reuse an existing solution?” 😁

r-Combination with repetition allowed

Example 6: Write a complete list to find the multisets of size 3 that can be selected from {1, 2, 3, 4}.

All combinations with 1, 1:	[1, 1, 1]; [1, 1, 2]; [1, 1, 3]; [1, 1, 4]
All additional combinations with 1, 2:	[1, 2, 2]; [1, 2, 3]; [1, 2, 4]
All additional combinations with 1, 3:	[1, 3, 3]; [1, 3, 4]
All additional combinations with 1, 4:	[1, 4, 4]
All additional combinations with 2, 2:	[2, 2, 2]; [2, 2, 3]; [2, 2, 4]
All additional combinations with 2, 3:	[2, 3, 3]; [2, 3, 4]
All additional combinations with 2, 4:	[2, 4, 4]
All additional combinations with 3, 3:	[3, 3, 3]; [3, 3, 4]
All additional combinations with 3, 4:	[3, 4, 4]
All additional combinations with 4, 4:	[4, 4, 4]

20 3-combinations with repetition allowed.

r-Combination with repetition allowed

Example 6: Write a complete list to find the multisets of size 3 that can be selected from {1, 2, 3, 4}.

Consider the numbers 1, 2, 3, 4 in {1, 2, 3, 4} as categories and imagine choosing a total of 3 numbers from the categories with multiple selections from any category allowed.

	Category 1		Category 2		Category 3		Category 4
[1, 1, 1]:	xxx	|		|		|	
[1, 3, 4]:	x	|		|	x	|	x
[2, 4, 4]:		|	x	|		|	xx

Hence, we may write [1, 1, 1] as "xxx|||", [1,3,4] as "x||x|x" and [2,4,4] as "|x||xx".

This is the same as $\binom{6}{3}$ or **20**.

r-Combination with repetition allowed

Theorem 9.6.1 Number of *r*-combinations with Repetition Allowed

The number of *r*-combination with repetition allowed (multisets of size *r*) that can be selected from a set of *n* elements is:

$$\binom{r+n-1}{r}$$

This equals the number of ways *r* objects can be selected from *n* categories of objects with repetitions allowed.

Example 6: $n = 4, r = 3. \binom{r+n-1}{r} = \binom{3+4-1}{3} = \binom{6}{3} = \mathbf{20}$

r-Combination with repetition allowed

At a hawker centre there are 6 stalls selling local delights. If you are buying lunch for 20 guests, how many different selections can you make?

Chicken rice



Nasi lemak



Mee rebus





Ayam penyet



Laksa



Bak chor mee

$n = 6, r = 20.$

$$\binom{r+n-1}{r}$$

$$= \binom{20+6-1}{20}$$

$$= \binom{25}{20}$$

$$= \mathbf{53130}$$

r-Combination with repetition allowed

Example 7: How many solutions are there to the give equations?

(a) $x_1 + x_2 + x_3 = 20$, each x_i is a nonnegative integer.

$n = 3, r = 20.$

$$\binom{r + (n-1)}{r} = \binom{20+2}{20} = \binom{22}{20} = \frac{22 \cdot 21}{2} = \mathbf{231}$$

(b) $x_1 + x_2 + x_3 = 20$, each x_i is a positive integer.

Convert to: $y_1 + y_2 + y_3 = 17$, each y_i is a nonnegative integer.

$n = 3, r = 17.$

$$\binom{r + (n-1)}{r} = \binom{17+2}{17} = \binom{19}{17} = \frac{19 \cdot 18}{2} = \mathbf{171}$$

Which formula to use?

	Order Matters	Order Does Not Matter
Repetition Is Allowed	n^k	$\binom{k+n-1}{k}$
Repetition Is Not Allowed	$P(n, k)$	$\binom{n}{k}$

Pascal's Formula and Binomial Theorem

Read the slides.

Probability Axioms

Recall: a sample space is a set of all outcomes of a random process or experiment and that an event is a subset of a sample space.

Probability Axioms

Let S be a sample space. A probability function P from the set of all events in S to the set of real numbers satisfies the following axioms: For all events $A, B \in S$,

1. $0 \leq P(A) \leq 1$
2. $P(\emptyset) = 0$ and $P(S) = 1$
3. If A and B are disjoint events ($A \cap B = \emptyset$), then $P(A \cup B) = P(A) + P(B)$

Probability of the Complement of an Event

Probability of the Complement of an Event

If A is any event in a sample space S, then

$$P(\bar{A}) = 1 - P(A)$$

Example: You roll a fair die, what is the probability of not rolling a 3?



P(rolling a 3) = $\frac{1}{6}$

P(not rolling a 3) = $1 - \frac{1}{6} = \frac{5}{6}$

Note: Epps uses A^c to represent the complement of event A.
We will use $\bar{A}$ to be consistent with the notation used in sets.

Probability of a General Union of Two Events

Probability of a General Union of Two Events

If A and B are any events in a sample space S, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Example: A card is drawn from a standard deck.
What is the probability of drawing a heart or a 7?



$P(H)$ = P(drawing a heart) $= \frac{13}{52} = \frac{1}{4}$

$P(7)$ = P(drawing a 7) $= \frac{4}{52} = \frac{1}{13}$

$P(H \cap 7)$ = P(drawing a 7 of heart) $= \frac{1}{52}$

$P(H \cup 7)$ = P(drawing a heart or a 7) $= \frac{1}{4} + \frac{1}{13} - \frac{1}{52} = \frac{4}{13}$

Expected Value

Definition: Expected Value

Suppose the possible outcomes of an experiment, or random process, are real numbers $a_1, a_2, a_3, \cdots, a_n$ which occur with probabilities $p_1, p_2, p_3, \cdots, p_n$ respectively. The **expected value** of the process is

$$\sum_{k=1}^{n} a_k p_k = a_1 p_1 + a_2 p_2 + a_3 p_3 + \cdots + a_n p_n$$

Expected Value



Example 14: If you roll a fair die, what is the expected value?

Each of the 6 outcomes (1, 2, … 6) has the same probability of 1/6, hence the expected value is

$$\frac{1}{6}(1) + \frac{1}{6}(2) + \frac{1}{6}(3) + \frac{1}{6}(4) + \frac{1}{6}(5) + \frac{1}{6}(6)$$

$$= \mathbf{3.5}$$

Expected Value

Example 15: Suppose that 500,000 people pay $5 each to play a lottery game with the following prizes:

- *A grand prize of $1,000,000,*
- *10 second prizes of $1,000 each,*
- *1,000 third prizes of $500 each, and*
- *10,000 fourth prizes of $10 each.*

Each of the 500,000 lottery tickets has the same chance as any other of containing a winning lottery number, and so $p_k = \frac{1}{500000}$ for all k = 1, 2, 3, …, 500000.

What is the expected value of a ticket?

Let $a_1, a_2, a_3, \ldots, a_{500000}$ be the net gain for an individual ticket, where $a_1 = 999995$.

$a_2 = a_3 = \ldots = a_{11} = 995$ (the net gain for each of the 10 second prize tickets)

$a_{12} = a_{13} = \ldots = a_{1011} = 495$ (the net gain for each of the 1000 third prize tickets)

$a_{1012} = a_{1013} = \ldots = a_{11011} = 5$ (the net gain for each of the 10000 fourth prize tickets)

$a_{11012} = a_{11013} = \ldots = a_{500000} = -5$ (the remaining 488989 tickets lose $5)

Expected Value

Example 15: Suppose that 500,000 people pay $5 each to play a lottery game with the following prizes:

- *A grand prize of $1,000,000,*
- *10 second prizes of $1,000 each,*
- *1,000 third prizes of $500 each, and*
- *10,000 fourth prizes of $10 each.*

What is the expected value of a ticket?

The expected value of a ticket is therefore:

$$\sum_{k=1}^{500000} a_k p_k = \sum_{k=1}^{500000} \left(a_k \cdot \frac{1}{500000} \right) = \frac{1}{500000} \sum_{k=1}^{500000} a_k$$

$$= \frac{1}{500000} (999995 + 10 \cdot 995 + 1000 \cdot 495 + 10000 \cdot 5 + 488989 \cdot (-5)) = \mathbf{-1.78}$$

Linearity of Expected Value

The expected value of the sum of random variables is equal to the sum of their individual expected values, regardless of whether they are independent.

For random variables X and Y (which may be dependent),

$$E[X + Y] = E[X] + E[Y]$$

More generally, for random variables $X_1, X_2, \cdots, X_n$ and constants $c_1, c_2, \cdots, c_n$,

$$E\left[\sum_{i=1}^{n} c_i \cdot X_i\right] = \sum_{i=1}^{n} (c_i \cdot E[X_i])$$

Linearity of Expected Value



Example 16: Expected value of sum of two dice.

	2	3	4	5	6	7
	3	4	5	6	7	8
	4	5	6	7	8	9
	5	6	7	8	9	10
	6	7	8	9	10	11
	7	8	9	10	11	12

Credit: brilliant.org

Calculating the expected value of the sum of two dice using basic method is tedious.

Using linearity of expectation, the expected value for the sum of two dice = 3.5 + 3.5 = 7

Check: Let s be the sum of two dice values

$P(s=2)=\frac{1}{36};\ P(s=3)=\frac{2}{36};P(s=4)=\frac{3}{36};P(s=5)=\frac{4}{36};P(s=6)=\frac{5}{36};P(s=7)=\frac{6}{36};$

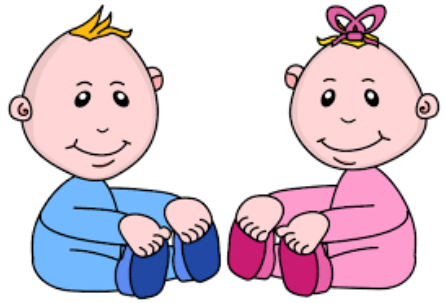
$P(s=8)=\frac{5}{36};P(s=9)=\frac{4}{36};P(s=10)=\frac{3}{36};P(s=11)=\frac{2}{36};\ P(s=12)=\frac{1}{36};$

Expected value = $2\cdot\frac{1}{36}+3\cdot\frac{2}{36}+4\cdot\frac{3}{36}+5\cdot\frac{4}{36}+6\cdot\frac{5}{36}+7\cdot\frac{6}{36}+8\cdot\frac{5}{36}+9\cdot\frac{4}{36}+$

$10\cdot\frac{3}{36}+11\cdot\frac{2}{36}+12\cdot\frac{1}{36}=7$

Conditional Probability

Imagine a couple with two children, each of whom is equally likely to be a boy or a girl. Now suppose you are given the information that one is a boy. What is the probability that the other child is a boy?



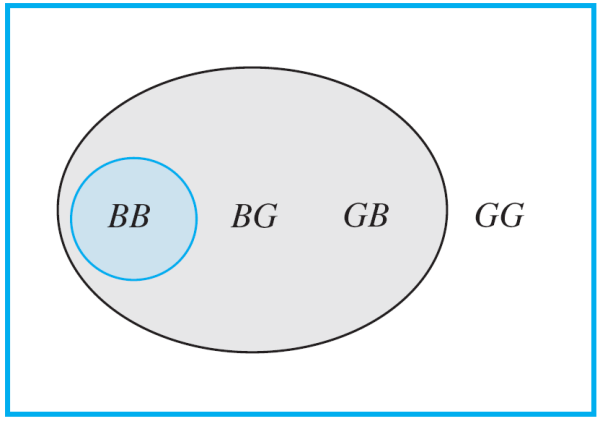


Figure 9.9.1

New sample space = gray region.

The likelihood that the other child is a boy given that at least one is a boy = 1/3.

Note also

$$\frac{P(\text{at least one child is a boy and the other child is also a boy})}{P(\text{at least one child is a boy})} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

Conditional Probability

A generalization of this observation forms the basis for the following definition.

Definition: Conditional Probability

Let A and B be events in a sample space S. If $P(A) \neq 0$, then the **conditional probability of B given A**, denoted $P(B|A)$, is

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \qquad 9.9.1$$

Multiplying both sides of formula 9.9.1 by $P(A)$, we get

$$P(A \cap B) = P(B|A) \cdot P(A) \qquad 9.9.2$$

Dividing both sides of formula 9.9.2 by $P(B|A)$, we get

$$P(A) = \frac{P(A \cap B)}{P(B|A)} \qquad 9.9.3$$

Conditional Probability



Example 17: An urn contains 5 blue and 7 gray balls. Let us say that 2 balls are chosen at random, one after the other, without replacement.

a. Find the following probabilities and illustrate them with a tree diagram: the probability that both balls are blue, the probability that the first ball is blue and the second is not blue, the probability that the first ball is not blue and the second ball is blue, and the probability that neither ball is blue.

b. What is the probability that the second ball is blue?

c. What is the probability that at least one of the balls is blue?

d. If the experiment of choosing two balls from the urn were repeated many times over, what would be the expected value of the number of blue balls?

Let

- S denote the sample space of all possible choices of two balls from the urn,
- B_1 be the event that the 1st ball is blue (then $\overline{B_1}$ is the event that the 1st ball is not blue),
- B_2 be the event that the 2nd ball is blue (then $\overline{B_2}$ is the event that the 2nd ball is not blue).

Conditional Probability



Example 17: An urn contains 5 blue and 7 gray balls. Let us say that 2 balls are chosen at random, one after the other, <u>without</u> replacement.

5 blue, 7 gray. a. Find $P(B_1 \cap B_2)$, $P(B_1 \cap \overline{B_2})$, $P(\overline{B_1} \cap B_2)$, $P(\overline{B_1} \cap \overline{B_2})$.

$P(B_1) = \frac{5}{12}$	$P(\overline{B_1}) = \frac{7}{12}$	$P(B_2 \mid B_1) = \frac{4}{11}$	$P(\overline{B_2} \mid B_1) = \frac{7}{11}$	$P(B_2 \mid \overline{B_1}) = \frac{5}{11}$	$P(\overline{B_2} \mid \overline{B_1}) = \frac{6}{11}$

By formula 9.9.2

$$P(A \cap B) = P(B|A) \cdot P(A)$$

$$P(B_1 \cap B_2) = P(B_2|B_1) \cdot P(B_1) = \frac{4}{11} \cdot \frac{5}{12} = \mathbf{\frac{20}{132}}$$

$$P(B_1 \cap \overline{B_2}) = P(\overline{B_2}|B_1) \cdot P(B_1) = \frac{7}{11} \cdot \frac{5}{12} = \mathbf{\frac{35}{132}}$$

$$P(\overline{B_1} \cap B_2) = P(B_2|\overline{B_1}) \cdot P(\overline{B_1}) = \frac{5}{11} \cdot \frac{7}{12} = \mathbf{\frac{35}{132}}$$

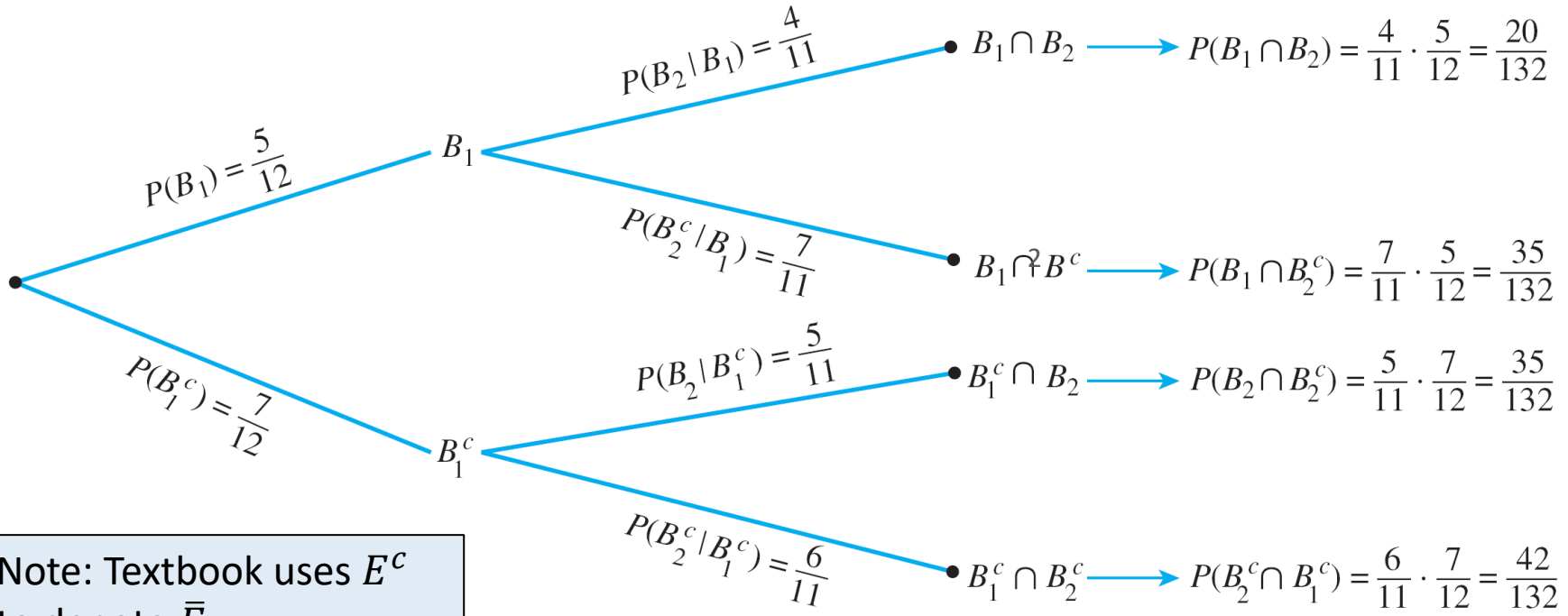
$$P(\overline{B_1} \cap \overline{B_2}) = P(\overline{B_2}|\overline{B_1}) \cdot P(\overline{B_1}) = \frac{6}{11} \cdot \frac{7}{12} = \mathbf{\frac{42}{132}}$$

Conditional Probability



Example 17: An urn contains 5 blue and 7 gray balls. Let us say that 2 balls are chosen at random, one after the other, <u>without</u> replacement.

5 blue, 7 gray. a. Find $P(B_1 \cap B_2)$, $P(B_1 \cap \overline{B_2})$, $P(\overline{B_1} \cap B_2)$, $P(\overline{B_1} \cap \overline{B_2})$.



Note: Textbook uses E^c to denote $\bar{E}$.

Figure 9.9.2

Conditional Probability



Example 17: An urn contains 5 blue and 7 gray balls. Let us say that 2 balls are chosen at random, one after the other, without replacement.

5 blue, 7 gray. b. What is the probability that the 2nd ball is blue?

First ball is blue and second is also blue, OR
first ball is gray and second is blue.
Note that they are mutually exclusive.

$$P(B_2) = P\big((B_2 \cap B_1) \cup (B_2 \cap \overline{B_1})\big)$$

$$= P(B_2 \cap B_1) + P(B_2 \cap \overline{B_1}) \quad \text{by probability axiom 3}$$

$$= \frac{20}{132} + \frac{35}{132} = \frac{55}{132} = \mathbf{\frac{5}{12}}$$

Conditional Probability



Example 17: An urn contains 5 blue and 7 gray balls. Let us say that 2 balls are chosen at random, one after the other, without replacement.

5 blue, 7 gray. c. What is the probability that at least one ball is blue?

Probability of a General Union of Two Events

If A and B are any events in a sample space S, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

$$P(B_1 \cup B_2) = P(B_1) + P(B_2) - P(B_1 \cap B_2)$$

$$= \frac{5}{12} + \frac{5}{12} - \frac{20}{132} = \frac{90}{132} = \mathbf{\frac{15}{22}}$$

Conditional Probability



Example 17: An urn contains 5 blue and 7 gray balls. Let us say that 2 balls are chosen at random, one after the other, without replacement.

5 blue, 7 gray.

d. The expected value of the number of blue balls.

P(no blue balls) = 1 – P(at least one blue ball) $= 1 - \frac{15}{22} = \frac{7}{22}$

The event that one ball is blue → first ball is blue but second ball is not; or second ball is blue but first ball is not.

From part (a), $P(B_1 \cap \overline{B_2}) = \frac{35}{132}$ and $P(\overline{B_1} \cap B_2) = \frac{35}{132}$

Hence, P(1 blue ball) $= \frac{35}{132} + \frac{35}{132} = \frac{70}{132}$

From part (a), P(2 blue balls) $= \frac{20}{132}$

Conditional Probability



Example 17: An urn contains 5 blue and 7 gray balls. Let us say that 2 balls are chosen at random, one after the other, without replacement.

5 blue, 7 gray. d. The expected value of the number of blue balls.

$P(\text{no blue balls}) = \frac{7}{22}$ $P(\text{1 blue ball}) = \frac{70}{132}$ $P(\text{2 blue balls}) = \frac{20}{132}$

Expected value of the number of blue balls

$= 0 \cdot P(\text{no blue balls}) + 1 \cdot P(\text{1 blue ball}) + 2 \cdot P(\text{2 blue balls})$

$$= 0 \cdot \frac{7}{22} + 1 \cdot \frac{70}{132} + 2 \cdot \frac{20}{132}$$

$$= \frac{110}{132} \cong \mathbf{0.8}$$

Bayes' Theorem

Conditional probablity:

$P(X) \neq 0$,

$$P(Y|X) = \frac{P(X \cap Y)}{P(X)}$$

$$P(X \cap Y) = P(Y|X) \cdot P(X)$$

Theorem 9.9.1 Bayes' Theorem

Suppose that a sample space S is a union of mutually disjoint events $B_1, B_2, B_3, \ldots, B_n$.

Suppose A is an event in S, and suppose A and all the B_i have non-zero probabilities.

If k is an integer with $1 \leq k \leq n$, then

$$\boldsymbol{P(B_k|A)} = \frac{\boldsymbol{P(A|B_k) \cdot P(B_k)}}{\underbrace{\boldsymbol{P(A|B_1) \cdot P(B_1) + P(A|B_2) \cdot P(B_2) + \cdots + P(A|B_n) \cdot P(B_n)}}_{P(A)}}$$

Bayes' Theorem



Suppose that one urn contains 3 blue and 4 gray balls and a second urn contains 5 blue and 3 gray balls. A ball is selected by choosing one of the urns at random and then picking a ball at random from that urn. If the chosen ball is blue, what is the **probability that it came from the first urn**?

This problem can be solved by carefully interpreting all the information that is known and putting it together in just the right way.

Let

- B be the event that the chosen ball is blue,
- U_1 the event that the ball came from the first urn, and
- U_2 the event that the ball came from the second urn.

Bayes' Theorem



3 of the 7 balls in the first urn are blue, 5 of the 8 balls in the second urn are blue:

The urns are equally like to be chosen: $P(U_1) = P(U_2) = \frac{1}{2}$

By formula 9.9.2 $P(A \cap B) = P(B|A) \cdot P(A)$

$P(B \cap U_1) = P(B\|U_1) \cdot P(U_1) = \frac{3}{7} \cdot \frac{1}{2} = \frac{3}{14}$
$P(B \cap U_2) = P(B\|U_2) \cdot P(U_2) = \frac{5}{8} \cdot \frac{1}{2} = \frac{5}{16}$

B is a disjoint union of $(B \cap U_1)$ and $(B \cap U_2)$, so by probability axiom 3,

$$P(B) = P\big((B \cap U_1) \cup (B \cap U_2)\big)$$

$$= P(B \cap U_1) + P(B \cap U_2) = \frac{3}{14} + \frac{5}{16} = \frac{59}{112}$$

Bayes' Theorem



3 of the 7 balls in the first urn are blue, 5 of the 8 balls in the second urn are blue:

From previous slide, $P(B \cap U_1) = \frac{3}{14}$ and $P(B) = \frac{59}{112}$

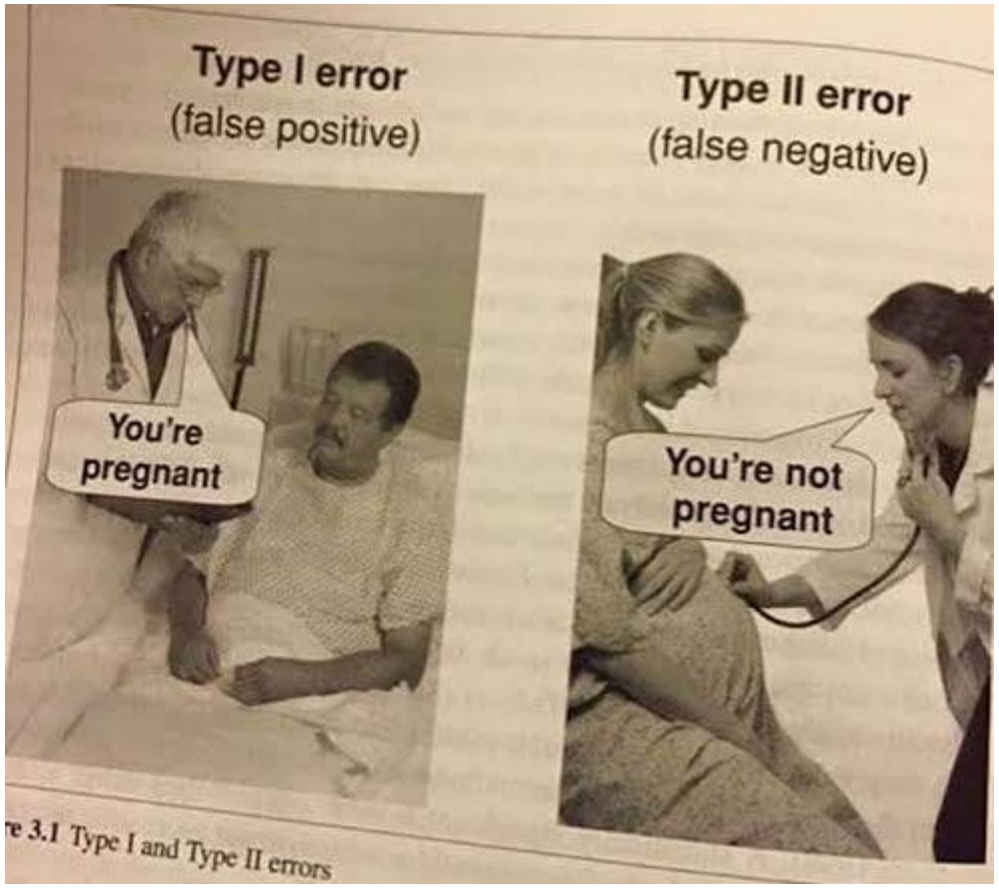
By definition of conditional probability,

$$P(U_1|B) = \frac{P(U_1 \cap B)}{P(B)} = \frac{\frac{3}{14}}{\frac{59}{112}} = \frac{336}{826} \cong 40.7\%$$

Thus, if the chosen ball is blue, the probability that it came from the first urn is approximately **40.7%**.

False Positive and False Negative

	Test tells you don't have it	Test tells you have it
You don't have it	True negative	False positive
You got it	False negative	True positive



Applying Bayes' Theorem

Example 18: Consider a medical test that screens for a disease found in 5 people in 1,000. Suppose that the false positive rate is 3% and the false negative rate is 1%.

Then 99% of the time a person who has the condition tests positive for it, and 97% of the time a person who does not have the condition tests negative for it.

a. What is the probability that a randomly chosen person who tests positive for the disease actually has the disease?

b. What is the probability that a randomly chosen person who tests negative for the disease does not indeed have the disease?

Applying Bayes' Theorem

Example 18

Consider a person chosen at random from among those screened.
Let

- A be the event that the person tests positive for the disease,
- B_1 the event that the person actually has the disease, and
- B_2 the event that the person does not have the disease.

Then $P(A|B_1) = 0.99$ $P(\bar{A}|B_2) = 0.97$

$P(\bar{A}|B_1) = 0.01$ $P(A|B_2) = 0.03$

Also, because 5 people in 1000 have the disease,

$P(B_1) = 0.005$ $P(B_2) = 0.995$

Applying Bayes' Theorem

Example 18

a. What is the probability that a randomly chosen person who tests positive for the disease actually has the disease?

$P(A|B_1) = 0.99 \quad P(\bar{A}|B_2) = 0.97$

$P(\bar{A}|B_1) = 0.01 \quad P(A|B_2) = 0.03$

$P(B_1) = 0.005 \quad P(B_2) = 0.995$

By Bayes' Theorem,

$$P(B_1|A) = \frac{P(A|B_1) \cdot P(B_1)}{P(A|B_1) \cdot P(B_1) + P(A|B_2) \cdot P(B_2)}$$

$$= \frac{(0.99) \cdot (0.005)}{(0.99) \cdot (0.005) + (0.03) \cdot (0.995)}$$

$$\cong 0.1422 \cong \mathbf{14.2\%}$$

Thus the probability that a person with a positive test result actually has the disease is approximately 14.2%.

Applying Bayes' Theorem

Example 18

b. What is the probability that a randomly chosen person who tests negative for the disease does not indeed have the disease?

$P(A|B_1) = 0.99$ $P(\bar{A}|B_2) = 0.97$

$P(\bar{A}|B_1) = 0.01$ $P(A|B_2) = 0.03$

$P(B_1) = 0.005$ $P(B_2) = 0.995$

By Bayes' Theorem,

$$P(B_2|\bar{A}) = \frac{P(\bar{A}|B_2) \cdot P(B_2)}{P(\bar{A}|B_1) \cdot P(B_1) + P(\bar{A}|B_2) \cdot P(B_2)}$$

$$= \frac{(0.97) \cdot (0.995)}{(0.01) \cdot (0.005) + (0.97) \cdot (0.995)}$$

$$\cong 0.999948 \cong \mathbf{99.995\%}$$

Thus the probability that a person with a negative test result does not have the disease is approximately 99.995%.

Past-Year's Exam Question!

AY2020/21 Sem1



In 1995, Mars© introduced the blue M&M's to replace the tan M&M's. Before that, the colour mix in a bag of plain M&M's was 30% brown, 20% yellow, 20% red, 10% green, 10% orange and 10% tan. After the replacement, the colour mix was 24% blue, 16% green, 20% orange, 14% yellow, 13% red and 13% brown.

Your friend Aiken, being a big fan of M&M's, has two bags of plain M&M's, one from 1994 and the other from 1996. He won't tell you which is which, but he gives you one M&M from each bag. One is yellow and one is green. What is the probability that the **yellow M&M came from the 1994 bag,** assuming the M&M's are picked randomly?

You may express your answer as a fraction or a decimal number correct to 3 significant digits.

Past-Year's Exam Question!

AY2020/21 Sem1



One is yellow and one is green. What is the probability that the **yellow M&M came from the 1994 bag,** assuming the M&M's are picked randomly?

Colour	1994	1996
Brown	30%	13%
Yellow	20%	14%
Red	20%	13%
Green	10%	16%
Orange	10%	20%
Tan	10%	-
Blue	-	24%

Let *A*: bag 1 from 1994 and bag 2 from 1996, and *B*: bag 1 from 1996 and bag 2 from 1994.

$$P(A) = P(B) = \frac{1}{2}$$

Let *E*: yellow from bag 1, green from bag 2.

$P(E|A) = 0.2 \times 0.16 = 0.032$ (yellow from 1994, green from 1996)

$P(E|B) = 0.14 \times 0.1 = 0.014$ (yellow from 1996, green from 1994)

By Bayes' Theorem,

$$P(A|E) = \frac{P(E|A) \cdot P(A)}{P(E|A) \cdot P(A) + P(E|B) \cdot P(B)} = \frac{0.032 \times \frac{1}{2}}{0.032 \times \frac{1}{2} + 0.014 \times \frac{1}{2}} =$$

$$\frac{0.016}{0.023} = \frac{\mathbf{16}}{\mathbf{23}} \text{ or } \mathbf{0.696}.$$

Independent Events

A is the event that a head is obtained on the first toss and *B* is the event that a head is obtained on the second toss, then if the coin is tossed randomly both times, events *A* and *B* should be ***independent*** in the sense that $P(A|B) = P(A)$ and $P(B|A) = P(B)$.

Observe that:

If $P(B) \neq 0$ and $P(A|B) = P(A)$,
then $P(A \cap B) = P(A|B) \cdot P(B) = P(A) \cdot P(B)$

By the same argument,

If $P(A) \neq 0$ and $P(B|A) = P(B)$,
then $P(A \cap B) = P(B|A) \cdot P(A) = P(A) \cdot P(B)$

Conversely,

If $P(A \cap B) = P(A) \cdot P(B)$ and $P(A) \neq 0$, then $P(B|A) = P(B)$.

If $P(A \cap B) = P(A) \cdot P(B)$ and $P(B) \neq 0$, then $P(A|B) = P(A)$.

Independent Events

Thus, for convenience and to eliminate the requirement that the probabilities be nonzero, we use the following product formula to define independent events.

Definition: Independent Events

If A and B are events in a sample space S, then A and B are **independent**, if and only if,

$$P(A \cap B) = P(A) \cdot P(B)$$

Independent Events

Let
- E be the event that a head is obtained on the first toss
- F be the event that a head is obtained on the second toss

$P(E) = P(F) = 0.6$.

Example 19: A coin is loaded so that the probability of heads is 0.6. Suppose the coin is tossed twice.

a. What is the probability of obtaining two heads?

$P(\text{2 heads}) = P(E \cap F) = P(E) \cdot P(F) = (0.6)(0.6) = 0.36 =$ **36%**

b. What is the probability of obtaining one head?

$P(\text{1 head}) = P\big((E \cap \bar{F}) \cup (\bar{E} \cap F)\big) = P(E) \cdot P(\bar{F}) + P(\bar{E}) \cdot P(F)$

$= (0.6)(0.4) + (0.4)(0.6) = 0.48 =$ **48%**

c. What is the probability of obtaining no heads?

$P(\text{no heads}) = P(\bar{E} \cap \bar{F}) = P(\bar{E}) \cdot P(\bar{F}) = (0.4)(0.4) = 0.16 =$ **16%**

Independent Events

Example 19: A coin is loaded so that the probability of heads is 0.6. Suppose the coin is tossed twice.

Let
- E be the event that a head is obtained on the first toss
- F be the event that a head is obtained on the second toss

$P(E) = P(F) = 0.6$.

d. What is the probability of obtaining at least one head?

Method 1:

$P(\text{at least one head}) = P(1 \text{ head}) + P(2 \text{ heads}) = 0.48 + 0.36$
$= 0.84 =$ **84%**

Method 2:

$P(\text{at least one head}) = 1 - P(\text{no heads}) = 1 - 0.16 = 0.84 =$ **84%**

Pairwise Independent/Mutually Independent

We say three events *A*, *B*, and *C* are *pairwise independent* if, and only if,

$P(A \cap B) = P(A) \cdot P(B)$, $P(A \cap C = P(A) \cdot P(C)$ and

$P(B \cap C) = P(B) \cdot P(C)$

Events can be pairwise independent without satisfying the condition

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C).$$

Conversely, they can satisfy the condition
$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$ without being pairwise independent.

Pairwise Independent/Mutually Independent

Definition: Pairwise Independent and Mutually Independent

Let *A*, *B* and *C* be events in a sample space *S*. *A* , *B* and *C* are **pairwise independent**, if and only if, they satisfy conditions 1 – 3 below. They are **mutually independent** if, and only if, they satisfy all four conditions below.

1. $P(A \cap B) = P(A) \cdot P(B)$
2. $P(A \cap C) = P(A) \cdot P(C)$
3. $P(B \cap C) = P(B) \cdot P(C)$
4. $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

Pairwise Independent/Mutually Independent

The definition of mutual independence for any collection of n events with $n \geq 2$ generalizes the two definitions given previously.

Definition: Mutually Independent

Events $A_1, A_2, \ldots, A_n$ in a sample space S are **mutually independent** if, and only if, the probability of the intersection of any subset of the events is the product of the probabilities of the events in the subset.

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1) \cdot P(A_2) \cdot \cdots \cdot P(An)$$

Pairwise Independent/Mutually Independent

Example 20: A coin is loaded so that the probability of heads is 0.6. Suppose the coin is tossed ten times.

a. What is the probability of obtaining eight heads?

$P(\text{eight heads}) = \binom{10}{8} (0.6)^8(0.4)^2$

b. What is the probability of obtaining at least eight head?

$P(\text{at least 8 heads}) = P(8 \text{ heads}) + P(9 \text{ heads}) + P(10 \text{ heads})$

$= \binom{10}{8} (0.6)^8(0.4)^2 + \binom{10}{9} (0.6)^9(0.4) + \binom{10}{10} (0.6)^{10}$

$\cong 0.167 =$ **16.7%**

Probabilities of the form $\binom{n}{k}p^{n-k}(1-p)^k$, where $0 \leq p \leq 1$, are called **binomial probabilities**.

Past-Year's Exam Question!

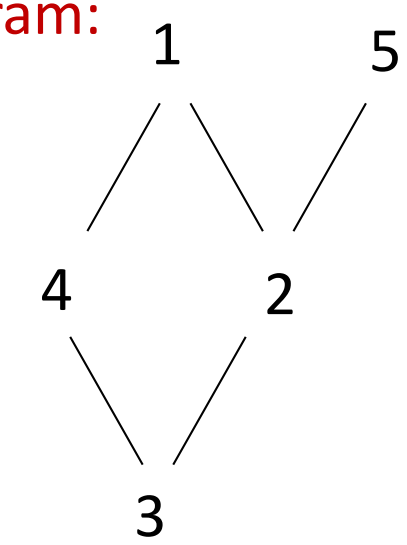
AY2021/22 Sem2 Q8

Given $A = \{1,2,3,4,5\}$ and the partial order R on A as follows:

$$R = \{(x,x) : x \in A\} \cup \{(2,1),(2,5),(3,1),(3,2),(3,4),(3,5),(4,1)\}$$

How many distinct linearizations of R are there?

Hasse diagram:



Linearizations:

5	1	5	1	1
|	|	|	|	|
1	5	1	5	4
|	|	|	|	|
2	2	4	4	5
|	|	|	|	|
4	4	2	2	2
|	|	|	|	|
3	3	3	3	3

5

Past-Year's Exam Question!

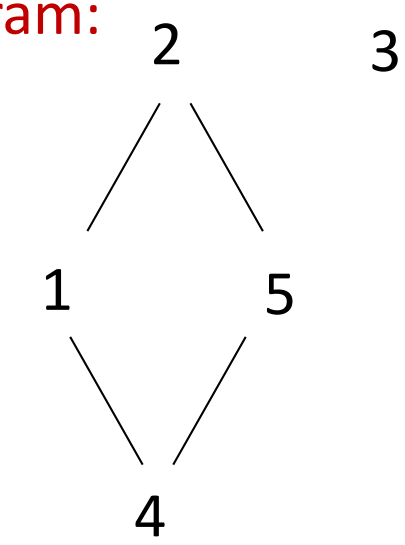
AY2022/23 Sem1 Q11

Given $A = \{1,2,3,4,5\}$ and the partial order R on A as follows:

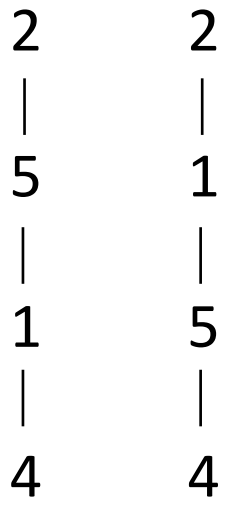
$$R = \{(x, x) : x \in A\} \cup \{(4,1), (5,2), (4,2), (1,2), (4,5)\}$$

How many distinct linearizations of R are there?

Hasse diagram:



Linearizations without 3:



3 can be placed in any of the 5 positions.

$2 \times 5 =$ **10**

Past-Year's Exam Question!

AY2022/23 Sem1 Q22(c)



Aiken rolled a fair 6-sided die twice. What is the probability that his second roll has a higher value than his first roll?

Write your answer as a simple fraction.

$$P(1)\cdot P(>1)+P(2)\cdot P(>2)+P(3)\cdot P(>3)+P(4)\cdot P(>4)+P(5)\cdot P(>5)+P(6)\cdot P(>6)$$

$$=\frac{1}{6}\left(\frac{5}{6}+\frac{4}{6}+\frac{3}{6}+\frac{2}{6}+\frac{1}{6}+\frac{0}{6}\right)=\frac{15}{36}=\mathbf{\frac{5}{12}}$$

Alternatively: P(both rolls have different values)

= 1 – P(both rolls have same value)

$$= 1-\frac{6}{36}=\frac{5}{6}$$

P(second value > first value) = P(first value > second value) (by symmetry)

Therefore, P(second value > first value) $=\frac{1}{2}\times\frac{5}{6}=\mathbf{\frac{5}{12}}$

Past-Year's Exam Question!

AY2022/23 Sem2 Q23(a)

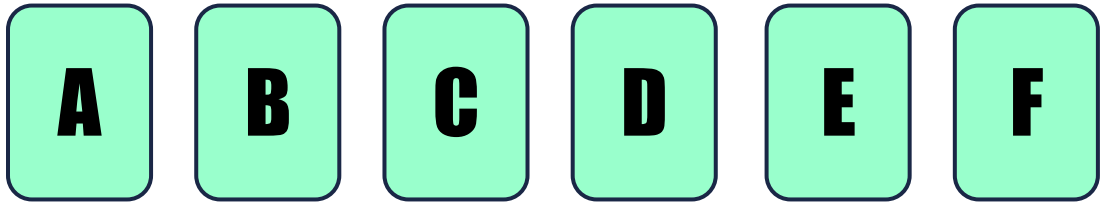


Aiken has thrown n fair 6-sided dice simultaneously, where $n > 0$. What is the probability that all n dice have the same number of dots on the faces showing up?

$$\frac{6}{6^n}$$

You have 6 distinct cards labelled *A*, *B*, *C*, *D*, *E*, *F*. They are laid face down in a row, but you know that *A* must be to the left of *B*, though they need not be next to each other.

How many possible arrangements are there?



Total arrangements = $6! = 720$

Half of them have *A* to the left of *B*, hence answer = $\frac{720}{2} = \mathbf{360}$.

You have 3 red cards (♥) and 3 black cards (♣), all distinct (e.g., A♥, 2♥, 3♥, A♣, 2♣, 3♣).

They are placed face down in a row with alternating colours.

How many arrangements are possible?



Either Red-Black-Red-Black-Red-Black or Black-Red-Black-Red-Black-Red

For each case, $3!\times3! = 36$

Answer $= 2\times36 = \mathbf{72}$

You have 12 distinct cards, comprising 4 suits and 3 ranks per suit.

How many possible arrangements are there if you place the cards in a row such that cards of the same suit stay together?



$$4! \times (3!)^4 = \mathbf{31104}$$

How many 4-digit even numbers are there if the digits are not repeated?

_ _ _ _

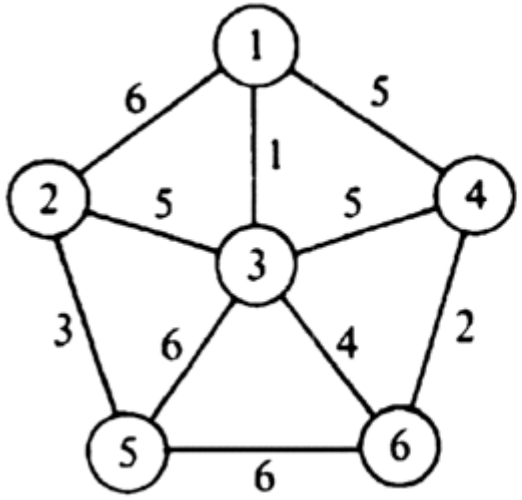
Case 1: Last digit = 0
Then there are $9 \times 8 \times 7 = 504$ even numbers.

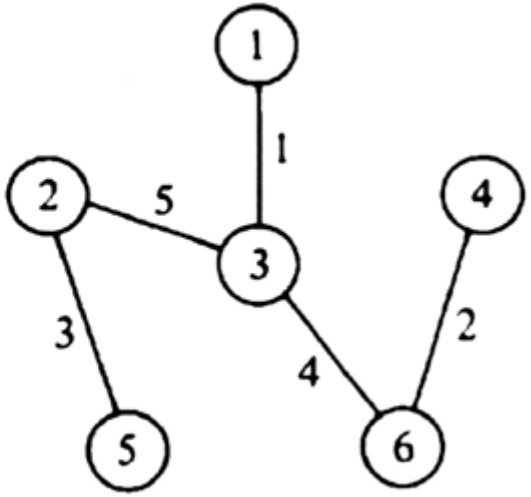
Case 2: Last digit $\in \{2,4,6,8\}$
Then there are $8 \times 8 \times 7 = 448$ even numbers.

Total $= 504 + (4 \times 448) =$ **2296** even numbers.

Next week's lectures

Graphs and Trees





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